SHRI RAMSWAROOP MEMORIAL UNIVERSITY

End Semester Examination (2023-24)-Odd Semester

B. Tech (CE)/ B. Tech (ME)/ B. Tech (EC)/ B. Tech (EE)/ B. Tech (EE) (EV)/ B. Tech (Bio-Tech)/ B. Tech (CSE) (CC & AI)/ B. Tech (CSE) (DS & AI)/ Integrated B. Tech (Bio-Tech)-M. Tech (Bio-Tech)/ Integrated B. Tech (Bio-Tech) - MBA– I Year (I Sem)

Course Name: Chemistry	**Code: UCY1801/BCY1701**
Time: 03 Hours	**Max Marks: 60**

University Roll No. | | | | | | | | | | | | | | | | |

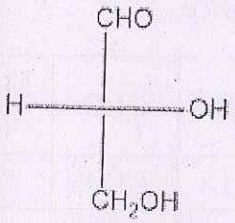
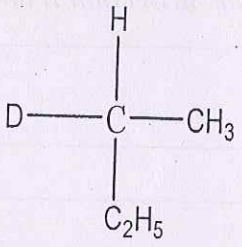
(To be filled by the Student)

Note: Please read instructions carefully:

a) *The question paper has 03 sections and it is compulsory to attempt all sections.*

b) *All questions of Section A are compulsory; questions in Section B and C contain choice.*

Section A: Very Short Answer type Questions **Attempt all the questions.**		**BL**	**CLO**	**Marks (10)**
1.	What causes temporary hardness in water?	BL1	CLO3	01
2.	Find the oxidation number of Fe in Fe_2O_3.	BL1	CLO3	01
3.	Explain bathochromic shift.	BL2	CLO2	01
4.	Show the ascending order of energy of the following electronic transitions: (i) π-π^* (ii) σ-σ^* (iii) n-σ^* (iv) n-π^*	BL2	CLO2	01
5.	First ionization potential of B is less than that of Be. Explain	BL2	CLO1	01
6.	Name the type of intermolecular force found in water and alcohol.	BL1	CLO3	01
7.	Discuss the term free energy with its formula.	BL2	CLO3	01
8.	What are nucleophiles? Give two examples.	BL1	CLO4	01
9.	Summarize elements of symmetry.	BL2	CLO4	01
10.	Explain, which type of disease is cured by phenacetin.	BL2	CLO4	01
Section B: Short Answer Type Questions **Attempt any 08 out of 09 questions.**		**BL**	**CLO**	**Marks (20)**
1.	Identify the ^{1}H-NMR signals in the following compounds and give reason for your answer: a. CH_3-CH_2-O-CH_2-CH_3 b. CH_3-CH_2- CH_2Cl	BL3	CLO2	2.5
2.	Explain chromophore with suitable examples.	BL2	CLO2	2.5
3.	Organize the following chemical species in ascending order of their size: (i) Na^+ (ii) Mg (iii) Cl (iv) Ar	BL3	CLO1	2.5
4.	Show the crystal field splitting and find the number of unpaired electrons in $[Ni(CO)_4]$.	BL3	CLO3	2.5
5.	Construct the molecular orbital diagram of N_2^+ and calculate its bond order.	BL3	CLO3	2.5
6.	Compare between bonding and anti-bonding molecular orbitals.	BL4	CLO3	2.5

Classify the following as soft or hard bases and justify your answer: NH_3, ROH, I^-, CO, O_2^-	BL4	CLO1	2.5
Illustrate E/Z configuration of isomers with an example.	BL4	CLO4	2.5
Discuss the synthesis of paracetamol with its pharmacological uses.	BL2	CLO4	2.5
tion C: Long Answer Type Questions **empt any 05 out of 06 questions.**	**BL**	**CLO**	**Marks (30)**
Summarize Bronsted-Lowry concept of acids and bases with examples.	BL5	CLO3	06
Discuss crystal field theory in detail. Determine the Crystal field stabilization energy (CFSE) and magnetic moment of $[Fe(CN)_6]^{3-}$.	BL6	CLO3	06
Illustrate the vibrational modes of molecules in Infrared spectroscopy.	BL4	CLO2	06
Elaborate the critical phenomena in Chemistry and explain the terms critical temperature, critical pressure and critical volume.	BL6	CLO3	06
Discuss the *cis-trans* isomerism and determine the *R/S* configuration of the following organic molecules:  (a)  (b)	BL5	CLO4	06
Outline the following terms with examples: a) Electronegativity b) Ionisation energy	BL4	CLO1	06